Lab 8: Using Pivot and Datasets in Splunk

Lab Overview

In this lab, you will learn to use Splunk's Pivot interface to create visualizations without writing complex searches. You'll build reports for business users using the drag-and-drop Pivot interface, create datasets for reusable analysis, and understand how Pivot transforms raw data into meaningful insights.

Learning Objectives

By the end of this lab, you will be able to:

- Access and use the Instant Pivot feature from search results
- Build reports using the Pivot interface with filters and splits
- Create visualizations using Split Rows and Split Columns
- Apply multiple filters to refine data analysis
- Save Pivot reports as dashboard panels
- Create and manage datasets from search results
- Navigate between Datasets, Pivot, and Search interfaces
- Understand the relationship between Pivot and data models

Prerequisites

- Completed Labs 1-6
- Understanding of basic Splunk search commands
- Familiarity with creating dashboards
- Basic knowledge of data visualization concepts

> **Data window**: The course dataset covers roughly **26 days** ending at the moment `labs/data/generate_course_data.py` was last run — deliberately sized to sit inside Splunk's **Last 30 days** preset. Set the time picker to **Last 30 days** for every exercise unless a step says otherwise.
>
> Regenerate the data within **4 days** of delivering the class; past that the preset starts clipping the oldest events and the answer-key counts drift.

Lab Environment Setup

Step 1: Verify Sample Data

Ensure required sample data is loaded:

```
index=main earliest=-30d latest=now
| stats count by sourcetype
```

Required sourcetypes:

- `access_combined_wcookie` - Web application logs with referrer data
- `db_audit` - Database audit logs
- `linux_secure` - Linux security logs

Step 2: Verify Dashboard

Ensure the Sales Dashboard from Lab 7 exists:

1. Navigate to **Dashboards**
2. Look for "Sales Dashboard"
3. If missing, create a new dashboard with this name

Key Concepts

Pivot Interface

What is Pivot?

- Visual interface for creating reports without SPL knowledge
- Drag-and-drop functionality for non-technical users
- Automatically generates underlying searches
- Built on top of data models for performance

Pivot Components

Component	Purpose	Example
Filters	Limit data to specific criteria	file=cart.do
Split Rows	Group data by field values (Y-axis)	productId
Split Columns	Create data series (X-axis)	referer_domain
Values	Metrics to calculate	Count, Sum, Average

Datasets vs Data Models

Datasets

- Simplified data models for specific use cases
- Created from search results
- Easier to manage than full data models

- Good for departmental reporting

Data Models

- Enterprise-wide data structures
- Support complex relationships
- Better performance for large-scale reporting
- Require more planning and maintenance

Exercise 1: Customer Referral Analysis

Scenario: The CFO wants to know where customers are coming from when they add items to their shopping cart.

Task 1.1: Access Instant Pivot

Start with a basic search:

```
index=main sourcetype=access_combined_wcookie
```

1. Run the search for **Last 30 days**
2. Click the **Visualization** tab
3. Observe three icons:
    - **Pivot** - Build tables and visualizations
    - **Quick Reports** - Pre-built report templates
    - **Search Command** - Return to search

Task 1.2: Launch Pivot Interface

1. Click the **Pivot** icon
2. In the modal window:
    - Select **All Fields**
    - Click **OK**

✓ **Checkpoint**: Pivot interface opens with field list on left

Task 1.3: Filter for Cart Activity

Under **Filters**:

1. Click **+** to add a filter
2. Select **file** from the Fields list
3. Configure filter:

- **Filter Type**: Match
- **Match**: is → `cart.do`

4. Click **Add To Table**

Expected Result: Table shows only cart.do events

Task 1.4: Add Product Breakdown

Under **Split Rows**:

1. Click **+** to add split rows
2. Select **productId** field
3. Configure:
   - **Label**: `Product Added To Cart`
   - Keep other defaults
4. Click **Add To Table**

✓ **Checkpoint**: Products appear as rows with counts

Task 1.5: Add Referrer Analysis

Under **Split Columns**:

1. Click **+** to add split columns
2. Select **referer_domain** field
3. Keep default settings
4. Click **Add To Table**

Observation: Notice buttercupgames.com dominates the traffic

Task 1.6: Filter Out Internal Traffic

Add another filter to exclude internal traffic:

Under **Filters**:

1. Click **+** to add another filter
2. Select **referer_domain** from Fields list
3. Configure:
   - **Filter Type**: Match
   - **Match**: is not → `http://www.buttercupgames.com`
4. Click **Add To Table**

Expected Results: External referrer domains now visible

Task 1.7: Change Visualization

1. Use the black sidebar on the left
2. Select **Line Chart** visualization
3. Observe the trend visualization

✓ **Checkpoint**: Line chart shows product additions by external referrer over time

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Exercise 2: Save to Dashboard and Create Dataset

Task 2.1: Save as Dashboard Panel

1. Click **Save As** → **Dashboard Panel**
2. Notice the data model fields:
   - **Model Title**: (Currently empty - will create new)
   - **Model ID**: (Will be auto-generated)
3. Configure:
   - **Dashboard**: Existing
   - **Dashboard Title**: Sales Dashboard
   - **Panel Title**: `Sales By Referral Domain`
   - **Model Title**: `Web Application Dataset`
   - **Model ID**: `web_app_ds`
4. Click **Save**
5. Click **View Dashboard**

✓ **Checkpoint**: New panel added to Sales Dashboard

Task 2.2: Access Your Dataset

1. Click **Datasets** menu on the top bar
2. Click **Yours** filter to show only your datasets
3. Find "Web Application Dataset" in the list
4. From the **Actions** menu, select **Explore** → **Visualize with Pivot**

✓ **Checkpoint**: Dataset opens in Pivot interface

Task 2.3: Explore Dataset Capabilities

Using your dataset, create a new analysis:

1. **Clear** any existing configuration
2. Add Filter:

- Field: **status**
- Match: is → `200`

3. Split Rows:
   - Field: **file**
   - Label: `Page Type`
4. Split Columns:
   - Field: **_time**
   - Period: Auto
5. Select **Column Chart** visualization

Exercise 3: Advanced Pivot Analysis

Scenario: Create a comprehensive traffic analysis showing user behavior patterns.

Task 3.1: Browser Usage Analysis

Starting from Pivot with your dataset:

1. **Filters**:
   - file → is → `success.do`
2. **Split Rows**:
   - useragent → Label: `Browser`
3. **Split Columns**:
   - productId → Limit: 5
4. Select **Pie Chart** visualization

Task 3.2: Geographic Analysis

Create location-based insights:

1. **Filters**:
   - status → is → `200`
2. **Split Rows**:
   - clientip → Label: `Client Location`
3. **Values**:
   - Count (default)
   - Add: Sum of bytes → Label: `Bandwidth`
4. Select **Statistics Table**

Task 3.3: Time-Based Patterns

Analyze traffic patterns over time:

1. **Split Columns**:
   - _time → Period: 1 day
2. **Split Rows**:
   - action → Label: `User Action`
3. Select **Area Chart** visualization

Challenge Exercises

Challenge 1: Multi-Dimensional Analysis

Create a pivot report showing:

- Products purchased (rows)
- By referrer domain (columns)
- Filtered for successful transactions only
- Excluding internal traffic
- As a heat map visualization

Challenge 2: Performance Dashboard

Build a complete dashboard using only Pivot:

1. **Panel 1**: Top browsers by traffic volume
2. **Panel 2**: Failed transactions by error code
3. **Panel 3**: Bandwidth consumption by file type
4. **Panel 4**: User sessions by hour of day

Challenge 3: Dataset Optimization

Create an optimized dataset:

1. Start with a filtered search (status=200)
2. Include only necessary fields
3. Add calculated fields for session duration
4. Save as a new dataset for team use

Working with Datasets

Dataset Management

Action	Purpose	Access Point
Create	Build from search results	Save As → Dataset
Edit	Modify fields and filters	Datasets → Manage
Share	Set permissions	Datasets → Permissions
Delete	Remove unused datasets	Datasets → Actions → Delete

Best Practices

1. **Naming Conventions**:
   - Use descriptive names
   - Include department/team prefix
   - Version datasets when updating
2. **Field Selection**:
   - Include only necessary fields
   - Add calculated fields in dataset
   - Document field purposes
3. **Performance**:
   - Limit time range where possible
   - Use acceleration for large datasets
   - Regular maintenance and cleanup

Pivot Interface Reference

Filter Options

Filter Type	Use Case	Example
Match	Exact value matching	file = cart.do
Does not match	Exclusion	status != 200
Contains	Partial string match	url contains "admin"
Is null	Missing values	referrer is null
In range	Numeric ranges	bytes 1000-5000

Split Options

Split Type	Purpose	Best For
Split Rows	Y-axis grouping	Categories, products
Split Columns	X-axis series	Time, locations
Split by	Color coding	Status, severity

Visualization Selection Guide

```
Data Pattern → Recommended Visualization
├── Trends over time → Line/Area Chart
├── Category comparison → Column/Bar Chart
├── Part of whole → Pie/Donut Chart
├── Multi-dimensional → Heat Map
├── Statistical summary → Statistics Table
└── Single metric → Single Value
```

Troubleshooting Guide

Common Issues and Solutions

Issue	Solution
No data in Pivot	Check time range and filters
Fields not appearing	Ensure "All Fields" is selected
Visualization not updating	Click "Apply" after changes
Cannot save dataset	Check write permissions
Pivot running slowly	Consider data model acceleration

Performance Tips

1. **Use Filters First**: Apply filters before splits to reduce data
2. **Limit Split Values**: Use "Limit" option for large cardinality fields
3. **Time Range**: Use specific time ranges instead of "All time"
4. **Dataset Acceleration**: Enable for frequently used datasets

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Cleanup

To clean up after this lab:

1. **Remove Dataset**:
    - Navigate to Datasets
    - Find "Web Application Dataset"
    - Select Delete from Actions menu
2. **Remove Dashboard Panel** (optional):
    - Open Sales Dashboard
    - Click Edit
    - Remove "Sales By Referral Domain" panel
    - Save dashboard

Key Takeaways

✅ **You've learned to:**

- Use Instant Pivot from search results
- Build complex reports without writing SPL
- Create and manage datasets for reusable analysis
- Apply multiple filters and splits in Pivot
- Choose appropriate visualizations for different data patterns
- Save Pivot reports to dashboards
- Navigate between Datasets and Pivot interfaces

Next Steps

- Proceed to Lab 9: Creating and Using Lookups
- Explore data model creation and acceleration
- Learn about Pivot table calculations and formatting

Quick Reference

Pivot Workflow

1. **Access Pivot**:
    - From Search: Visualization tab → Pivot
    - From Dataset: Datasets → Visualize with Pivot
    - From Data Model: Settings → Data models → Pivot

2. **Build Report**:
    - Add Filters → Define criteria
    - Add Splits → Group data
    - Select Values → Choose metrics
    - Pick Visualization → Display results
3. **Save Options**:
    - Save as Report
    - Save as Dashboard Panel
    - Save as Dataset (from search)

Common Pivot Patterns

Analysis Type	Filters	Split Rows	Split Columns
Product Sales	file=success.do	productId	_time
Traffic Sources	status=200	referer_domain	date_hour
Error Analysis	status>=400	status	file
User Behavior	-	action	clientip

Answer Key

🔑 **Instructor answer key**

Exercise 1

- Most cart additions come from: buttercupgames.com (3,432), google.com (2,941), facebook.com (1,410)
- Top products added to cart: CU-PG-G06, PZ-SG-G05, WC-SH-A02

Exercise 2

- Dataset created: web_app_ds
- Panel added: Sales By Referral Domain

Dataset Fields

- Key fields: file, productId, referer_domain, status, clientip
- Time field: _time